2

A Conceptual Framework for Productive Development Policies

Chapter 1 made the case that the relevant question is not whether productive development policies (PDPs) should be pursued; rather, the question is how. What types of policies can help address the many market failures that beset the process of productive development—without repeating the errors of the past—and pave the way for a better future? In this regard, the oft-cited image of a pendulum, by which past industrial policies have swung out of vogue and are now back in fashion, is simply not appropriate. It would be more accurate to say that certain well-designed policy interventions may be worthwhile; they have stood the test of time, or can be crafted to meet new challenges. Other types, however, may lack proper conceptual justification or may be too risky to implement effectively. What types of policies should be favored? Which ones should be set aside? There is a need to rethink PDPs, to screen them in order to separate the wheat from the chaff. Rethinking these policies starts with a proper conceptual framework.

What Types of Policies Are Needed?

PDPs differ in a variety of dimensions. This framework emphasizes two. The first dimension pertains to their scope. They can focus on specific sectors (vertical policies). Or they can be broad-based and not attempt to benefit any industry in particular (horizontal policies).[1]

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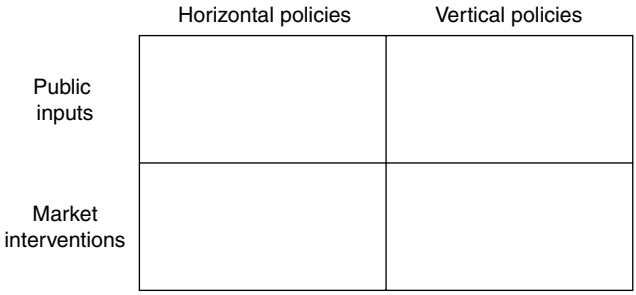
The second dimension relates to the type of intervention. Support can take the form of *public inputs* or *public goods* that the state can provide in order to enhance the competitiveness of the private sector, such as general improvements in infrastructure or protection of property rights. Or support can take the form of *market interventions*—such as subsidies, tax breaks, or tariffs—that affect the incentives faced by private actors, and thus influence their behavior. These two dimensions can be combined in a 2×2 matrix, which divides the universe of PDPs into four quadrants (Figure 2.1).

On the horizontal dimension, an example of a horizontal public input would be an intervention that reduces the transaction costs of starting a business. It is a public good provided by the state, and it is not specifically intended to benefit any particular sector. An example of a horizontal market intervention would be a matching grant scheme for R&D. On the vertical dimension, an example of vertical public input would be the provision of phytosanitary controls that prevent plagues from contaminating food products. An example of vertical market intervention would be giving tax breaks to a specific sector, such as tourism.

Why focus on these dimensions? For a simple but powerful reason: Because the public policy considerations to take into account when analyzing the merits of PDPs differ in each of these quadrants.

This is particularly true when it comes to the problems of rent seeking and capture precisely the type of problems that gave industrial policies a bad reputation in the past. Interventions in these quadrants differ importantly in the degree to which they are subject

Figure 2.1 A Typology of PDP Interventions



to these problems. Generally, these problems tend to be more pervasive in market interventions such as subsidies, tariff protection, or tax breaks than in the provision of public inputs. Even when the policy rationale calls for temporary measures, market interventions tend to produce ongoing benefits with open-ended time frames, and thus create strong constituencies that lobby for continued support. By contrast, public inputs supplement productive efforts by the private sector and usually have no reason to be discontinued, and thus do not face the same problems.

Rent-seeking problems are also likely to be more prevalent in the case of vertical interventions. As these policies generate concentrated benefits, and the stakes for beneficiaries tend to be higher, vertical interventions tend to create incentives for the sectors to lobby for support, and facilitate the necessary collective action at the sector level to sustain such lobbying.[2] Taking both the ongoing nature and the high stakes profile into account, interventions in the vertical/market intervention quadrant seem riskier from a political economy point of view, and thus require particularly careful safeguards (see Dal Bó and Finan, 2014).

The discussion that follows presents a bird's-eye view of the type of considerations that should be taken into account when thinking about the merits of policy interventions in each of these four quadrants.

Horizontal Public Inputs

This quadrant includes interventions that ensure respect for property rights, efforts to improve the quality of education or the general quality of the infrastructure, and measures to streamline the steps and costs of starting a business.

Of all the quadrants, this is the least controversial. Even the most ardent defenders of the free market would agree that the state has a central role to play in this type of policy. Many of the interventions in this quadrant are associated with the cost of doing business in a country, and many of them are captured by indexes such as the Doing Business Index of the World Bank, and the Global Competitiveness Index of the World Economic Forum. Governments around the

world have often made these indexes the focus of their competitiveness policies. Rather than analyzing the merits of state intervention, which are not under discussion, the key in this quadrant is to consider whether the particular policies pass three key tests: whether the design of the interventions is appropriate for achieving the desired objectives, whether these interventions have the desired impact, and whether they are cost-effective.

These policies are obviously very important, and they involve substantial challenges. But because they are less controversial, they are in less need of rethinking. For these reasons, the discussion of policies corresponding to this quadrant—mainly in this chapter, and to some extent, later in this book—will be left aside, in order to focus on interventions in the other quadrants.

Horizontal Market Interventions

These interventions are not aimed at favoring specific *sectors*, but rather at stimulating certain *activities*. Examples include tax breaks to attract foreign investment, subsidies for job training, for investment in machinery, or subsidized credit to small and medium enterprises (SMEs). Are such interventions justified by a sound rationale? This is the critical question in this quadrant. And the answer is . . . It depends.

The key is to identify the market failure that these interventions seek to address. A *market failure* implies that the net social benefit of productive activities differs from the net private benefit perceived by market agents that control these activities: that is, these productive activities generate spillover effects that may benefit or impose costs on other agents. In some cases, these market failures are clear. Consider a firm's decision to invest in training its workers. The firm faces the following problem: some of the trained workers may go on to work for other firms—perhaps even competing firms. Thus the firm cannot appropriate all the benefits associated with the training. In this case, there is a spillover: when deciding how much to invest, the firm will take into account only its own private benefits associated with the training, not the total benefits to society, part of which will be appropriated by others. As a result, the firm's investment in

training will be less than what would be desirable from the point of view of society as a whole (that is, the market fails to achieve a socially appropriate solution). In such cases, a subsidy to promote training can be justified, so that the firm has incentives to provide the socially optimal level of job training.[3] A similar case can be made for subsidizing R&D that has spillovers beyond the firm.

Compare this with the case of investment in machinery. Assume that the machine will increase the productivity of a firm. Is this a good enough reason for the state to intervene with a subsidy? The answer is no. To the extent that the firm can appropriate all the benefits associated with this increased productivity, the firm itself would have all the right incentives to invest without the need for a subsidy. Unless there are special circumstances that determine the existence of a market failure, state intervention would not be justified in this case.[4]

Once a market failure has been identified, a second important question is whether the intervention proposed or adopted is designed so as to address as precisely as possible the market failure that has been identified. For example, although an investment in research and development can clearly create positive spillovers, certain R&D activities are more likely to generate them than others. Collaborative R&D activities between groups of firms, or between firms and research institutions, tend to reduce duplication costs and create more spillovers than R&D activities within individual firms, which could have an interest in limiting dissemination of the knowledge acquired. Therefore, interventions designed to encourage collaboration in R&D projects have more chance of successfully addressing market failures than interventions that stimulate R&D regardless of how it is done. These and other aspects of the design of R&D incentives are discussed in detail in chapter 3 on innovation and technology diffusion.

A similar case can be made with respect to tax breaks to attract FDI. The justification for these exemptions is that there are positive spillovers associated with the installation of foreign companies in the host country. The literature, however, is inconclusive in this respect. Studies such as Aitken and Harrison (1999) suggest that spillovers could be negative, particularly within the sector, since new

companies compete with existing ones for productive resources, resulting in higher factor costs. In contrast, other studies suggest that there could be positive spillovers through linkages to local suppliers (provided that they have the human capital required to absorb the foreign company's technology), the development of specialized inputs, or the mobility of labor (if employees trained in the foreign firm then go on to work in other companies or start their own businesses).[5] Interventions aimed at fostering these spillovers (for example, tax breaks linked to certain labor training commitments, or a foreign firm's support for programs to develop local suppliers) can be preferable to interventions aimed at promoting foreign investment across the board.

A third important question has to do with the size of subsidies or exemptions. In general, the size of the subsidy should be related to the size of the spillover. In the job training example, unless turnover is very large, a substantial part of the training benefits is captured by the firm that is making the investment in training, in the form of the increased productivity of its workforce. In this case, a subsidy amounting to a small fraction of training costs might be warranted, but not a grant that covers the full cost of training.[6] In the case of exemptions or subsidies to attract foreign investment, the ideal would be to offer the minimum necessary for the investment to occur, as long as this minimum is less than the size of the spillover.[7] In this way, the host country—and not the foreign firm—can capture a larger share of the benefits associated with the new investment.

Another aspect that can be important in certain cases has to do with the timing of the benefits associated with certain market interventions. Consider the case of attracting FDI. While the installation of the firm in the host country is a one-time event, the instruments used to attract it may differ in terms of their timing. They could take the form of a one-time benefit, such as an installation grant, or a benefit that is extended over time, such as an exemption from certain taxes for a certain number of years. In addition to the potential issue that recurring benefits may be difficult to discontinue, another political economy issue may come into play. In the case of the tax exemption, most of the sacrifice in terms of fiscal resources may fall on future administrations, particularly if the investments do

not bear fruit for a few years. Current governments may accrue the immediate political benefits associated with extending tax breaks, without internalizing the costs. Thus incentives may be overly generous, with the corresponding cost to future taxpayers.[8]

In addition, interventions in this quadrant can also be used to stimulate the development of competitive new export activities. What Hausmann and Rodrik (2003) have called the process of "self-discovery" is of particular policy interest. According to these authors, the activity of exploring new sectors with potential comparative advantages occurs less often than it should because of the presence of spillovers. Developing a competitive new export activity—and in the process "discovering" that the country has comparative advantages in its production—requires costly experimentation. While the pioneer must cover these costs—and therefore will suffer the corresponding losses if the experiment does not pan out—success can give rise to imitation. Thus the successful pioneer cannot appropriate all the benefits associated with the "discovery" because some of the benefits will be enjoyed by the followers. This may call for a subsidy to stimulate discovery activities by compensating pioneers for this spillover. Box 2.1 presents a scheme to do precisely this, which also serves as an illustration of how policy instruments in this quadrant can be designed so as to address market failures in a precise way.

Box 2.1 Addressing the Self-Discovery Problem with Subsidies to Export Pioneers

The case of blueberries in Argentina is an excellent example of the type of externalities involved in self-discovery (see Sánchez et al., 2012). At the beginning of the 1990s, the pioneer Francisco Caffarena was searching for investment alternatives with export potential in the agricultural sector. Blueberries had the potential to get to market during the Northern Hemisphere off-season, when prices, and thus profits, were expected to be high.

Since blueberries were not produced in Argentina at the time, he faced uncertainty. Would the climate and soil conditions be favorable? What would be the ideal location within the country? What varieties of blueberry would be more appropriate for the local conditions? When would he

be able to harvest, and at what price would he be able to sell? To launch his business, Caffarena had to invest in experimentation. He hired US experts to advise him in matters of production and sanitation; he imported several different varieties of blueberry plants and learned to reproduce them; he planted different varieties in different locations. Altogether, the initial investment was approximately US$ 200,000.

He discovered that local conditions were appropriate for blueberry production. But in the process, he also discovered something of great value: he could harvest one month before anyone else in the world, thus avoiding competition and fetching very high prices. Not surprisingly, soon he had many followers. Furthermore, since Argentina as a source country did not have competition at the time of the early harvest, the presence of followers led to lower prices and eroded his profitability.

In response to these developments, the pioneer moved upstream (a nursery to supply plants to other producers) and downstream (export services to the followers). Predictably, in time, competition in these activities emerged as well. By 2005, his share of exports was less than 5 percent. Clearly, the pioneer was not able to appropriate all the benefits associated with his discovery. Diffusion generated a positive externality, benefiting followers and contributing to the country's export diversification.

In this case Caffarena, somehow, decided to invest. Whether or not he underestimated the extent to which he would be followed, he obviously thought that his expected private benefits would exceed the costs, which they did. Many other would-be pioneers, however, may be discouraged from experimenting. As a result, some potentially competitive products may be left undiscovered. What is the best way to address this market failure? Three potential interventions will be discussed:

i. Ex ante Subsidies for Self-Discovery

The idea, proposed by Rodrik (2004), is to identify potential pioneers and subsidize their investment in experimentation. But how can pioneers be identified ex ante? How can the process be protected from rent seeking and favoritism? Rodrik proposes a contest in which entrepreneurs present pre-investment proposals to compete for public funds. Eligible projects would relate to substantially new activities and demonstrate potential to generate spillovers. Selected projects would be subject to audits and oversight. Given that experimentation in self-discovery is risky, many projects would not pan out, and not generate spillovers. As Rodrik would put it, however, each success would compensate for many failures. But the question remains, is it possible to do better?

ii. The Simplified Drawback in Chile

In 1985, Chile implemented a simplified drawback, or *reintegro simplificado*. In lieu of reimbursements for tariffs paid on imported inputs (drawbacks), exporters received a subsidy of up to 10 percent of their export value, as long as total exports of the sector did not exceed $7.5 million.[9] Once this threshold was breached, export subsidies in the sector were automatically eliminated. One strength of this intervention is that it did not require picking winners. Subsidies were assigned automatically to every exporting firm in a new sector, thus making the development of new or incipient sectors more attractive to pioneers and early followers. Unlike Rodrik's proposal, those that experimented but failed would not receive support. There were, however, a few downsides. First, the subsidy would not just go to the pioneer who invested in experimentation, but to early followers as well. While experimentation was encouraged, the early imitation that exacerbates the appropriation problem was encouraged as well. Second, the successful pioneer was subsidized whether or not he generated spillovers, that is, whether or not he was followed. Third, given that the subsidy was eliminated as soon as the threshold was breached, the pioneer had incentives to discourage diffusion and thus minimize spillovers, since faster diffusion led to earlier phaseout. The *reintegro simplificado* was dismantled in 2003.

iii. Addressing the Self-Discovery Problem More Precisely: A New Policy Proposal

The idea, proposed by Stein (2012), is to simply subsidize export pioneers in proportion to the exports of their followers (thus not including their own exports). As in the case of the *reintegro simplificado*, picking winners is not required. Subsidies go automatically to those that generate a successful discovery as long as they are followed. The proposed scheme addresses the three weaknesses of the *reintegro simplificado*: (i) it benefits only the pioneer, not the followers; (ii) it benefits him only to the extent that he generates spillovers; and (iii) it generates the right incentives for the pioneer to encourage diffusion.[10] The subsidy would not need to be large. In the case of blueberries discussed above, for example, Stein (2012) estimates using Argentina's customs export data that a subsidy of 2 percent of exports of followers would have paid the pioneer about $1.2 million between 1998 and 2005, more than enough to compensate for the cost of experimentation.[11]

This program would be fully compatible with WTO rules for the case of agricultural products, such as the blueberry case discussed above, provided the parameters of the subsidy fall within the limits agreed upon

by the country in the Uruguay Round (see Box 1.2 in chapter 1). More generally, to the extent that this scheme is understood as an incentive for research and development, it would be compatible with the spirit of WTO restrictions. Furthermore, while WTO rules prohibit subsidies conditioned on export performance (with the exception of agriculture), under the proposed policy the subsidy would not be conditioned on own exports, only on those of others. Even if the subsidy were classified by the WTO within the "prohibited" or "actionable" categories, making it sufficiently small (below 2 percent) would ensure that it is not challenged, provided the export share of the country in the destination market is sufficiently low, owing to WTO special de minimis provisions for developing countries.[12] Larger subsidies may simply require this proposal to be adjusted so that it does not discriminate whether pioneers and followers export or produce for the domestic market.

In some cases, interventions in this quadrant involve the participation of agents that help connect the demand for public support with the corresponding supply, intermediating between the state and the beneficiary firms. Examples include first-tier commercial banks, which intermediate between second-tier state development banks and the private firms that receive their credit; incubators, which intermediate between a state agency dealing with entrepreneurship and the aspiring entrepreneurs that receive subsidized services from the incubator; and providers of different types of technological or managerial services, which intermediate between a development agency that promotes SMEs and the target beneficiaries. These intermediary agents may play a crucial role because they are closer to the clients and know their needs. They bring an element of flexibility and a private sector perspective into the selection of beneficiaries, and may act as screening devices, identifying opportunities that would otherwise go unnoticed given the more rigid structures of the public sector.[13]

At the same time, interventions provided through intermediary agents are subject to well-known principal-agent problems. The objectives of the intermediary agents are not necessarily aligned with those of the principal. Thus mechanisms may need to be put in place to align their incentives so that the development objectives can

be achieved.[14] In the case of incubators, for example, it is important that these intermediaries receive compensation on the basis of the success of incubated firms, rather than simply the number of firms subject to incubation. This provides them with the right incentives to select the right projects, and to add value during the incubation process. These issues are discussed further in chapter 4 on entrepreneurship, with a particular focus on a change in incubator policy in Chile.[15]

To summarize, when analyzing the merits of policy interventions in this quadrant—whether grants, tax breaks, subsidized credit, or other types—the key is to identify the market failure (if any) that justifies the intervention, and adjust the policy instruments to address the identified market failure as precisely as possible.

Vertical Public Inputs

This quadrant encompasses public inputs provided or arranged by the state that generate benefits for specific sectors. Examples in Latin America and the Caribbean include phytosanitary controls; the promotion of tourism from abroad; the creation of specific university degrees to attract a certain type of investment; research by public agencies to create new, more productive crop varieties; or coordination of critical aspects of logistics to facilitate export of specific products, such as cold chain logistics to allow the export of certain fruits and vegetables.

Policies in this quadrant are selective, and thus favor some sectors and not others. These policies are more controversial, since they involve the highly contentious issue of "picking winners," which became a lightning rod for criticism, particularly during the period of the backlash against industrial policy, and remains controversial today.

Should certain sectors be chosen to receive support? If so, how should these sectors be picked? How can a public-private dialogue be structured to identify the public inputs that are needed? How should the state organize itself in order to be able to deliver the required public inputs or help the private sector produce the collective inputs they need? How can problems of capture be avoided? As can be

appreciated, the important questions in this quadrant are very different from those of the last one.

It is worthwhile asking first if the state should provide public inputs that benefit specific sectors, or if instead it should focus solely on providing the type of public inputs that are horizontal in nature. The answer is an emphatic yes: the state should get involved in this quadrant. As Hausmann and Rodrik (2006) convincingly argue, most public inputs, by their very nature, tend to benefit specific sectors. Implementation of cold chain logistics can be a key input for the fruit and vegetable sector, but not for the textile sector; a good intellectual property law can be a key input for the biotechnology sector, but not for footwear; development of a university degree in mechanical engineering in the public university (as Costa Rica did to attract Intel) can be a key input for the electronics industry, but not for agriculture. To reserve state action only for those public inputs that have neutral effects on different sectors would mean leaving out a number of interventions that clearly contribute to the productivity of the economy or to the sound functioning of certain markets, which in most cases would not be provided spontaneously by the private sector.

While there is a clear case for the state to provide vertical public inputs, the public sector does not necessarily need to pay for them. In fact, many of these interventions are actually collective or club goods that benefit a specific set of firms, which ought to be willing to foot the bill. While in some cases sectors may be able to organize and provide the collective goods themselves, in others the state may have to help solve a coordination problem that may preclude atomized firms from providing their own solution. A case in point is discussed in Box 2.2, which describes efforts to promote rice production in the province of Entre Ríos in Argentina. More generally, policies in this quadrant should normally include provisions for the private sector to share at least some of the costs of financing the solutions to their problems. In fact, their willingness to pay for a public input is a key test of whether the intervention in question is money well spent. These issues will be discussed in more detail in chapter 11.[16]

If the state has a role in securing these public inputs, how should it choose which inputs to provide? Whose needs should be dealt with

first? Hausmann and Rodrik argue that the specific nature of the benefits generated by public inputs, combined with budgetary constraints and the limited attention span of policymakers, mean that policymakers are "doomed to choose." But that does not necessarily mean that the sectors need to be selected a priori. While some countries have preselected sectors—such as Chile, with its cluster program, and Colombia, with its Productive Transformation Program (PTP)—another possible strategy is to initiate a dialogue with any sector that organizes itself to ask for public input support, and then choose which of the problems are worth solving without defining in advance the sectors to be supported. The relative merits of these different strategies may depend on the characteristics of the country in question, a matter that will be discussed in more detail in chapter 9, which addresses the issue of sector selection in vertical policies in detail.

If governments do choose sectors a priori, what criteria should they use? An obvious criterion is to target export sectors in which world demand is growing and in which the country has demonstrated, or at least can be presumed to have, latent comparative advantage. Chapter 1 suggested that countries may also want to focus on sectors that are relatively sophisticated or complex, or that have strategic value in that they open avenues for later redeployment of productive capabilities to other worthwhile sectors. As with the discussion of the previous quadrant, however, choosing sectors with these criteria makes sense only to the extent that there are market failures that prevent the private sector from developing these promising sectors or allowing them to reach their potential. These issues will also be tackled in detail in chapter 9.

Regardless of the criteria used to select sectors, though, it is important to ensure that the selection mechanism is based on an in-depth analysis of the sector's challenges and opportunities and is largely free from political interference and from the vested interests of the sectors involved. For example, Chile heavily involved an international consulting firm, the Boston Consulting Group, in the selection of the sectors in its cluster program. It is also crucial that support be quickly abandoned if it becomes clear that the sectors do not have competitive potential.[17]

For the state to be able to help solve the problems holding back the development of a certain sector, it must be able to identify what those problems are. To carry out this diagnosis, the dialogue between the public and private sectors is absolutely essential. This dialogue often takes place within the context of cluster development programs (CDPs), studied in chapter 7. How the state structures this dialogue, and the nature of the conversations that take place within it, are key for the success of interventions in this quadrant. It is important that the sector-level dialogue focuses on the public inputs, the solutions to coordination problems, or the removal of obstacles required by the private sector to become more productive—and not on subsidies, tax breaks, or other market interventions that may improve profitability without making a dent in productivity. The public sector's role in shaping this dialogue is to allow the necessary exchange of information and support public-private collaboration, while limiting the likelihood of rent seeking. Chapter 11 addresses these issues of how best to structure public-private interaction for PDPs.

Once the priority actions have been identified, it is important to understand how to organize the public sector's response to the private sector's needs. A good response capacity requires good coordination within the public sector because the public agencies that identify needs in dialogue with the private sector are not necessarily the ones that will be responsible for providing the public inputs required to meet these needs. For example, the minister of tourism may identify the need to improve a road required to develop a tourist destination, but the agency that has the resources and the responsibility to deal with road construction and maintenance may be the Ministry of Public Works. Without a system in place that allows authority, information, and resources to flow within the public sector to deliver the necessary public inputs, identified inputs will not be delivered appropriately, and the private sector may be discouraged from participating in the sector-level dialogue.

The Productive Transformation Program in Colombia illustrates some of these points. In this program, a number of sectors—some of them emerging, some more traditional—have been selected with the aim of taking them to world-class status. In each of them, public-private collaboration has been structured to identify obstacles,

public inputs that are needed, and coordination problems, and come up with an action plan to address them, with the help of an international consulting firm. The rules of the program clearly delimit the nature of the conversations that can take place in the sector-level dialogue: there is no scope to discuss subsidies, protection, or other market interventions. While there has been some partial success in some sectors (for example, cosmetics), critics charge that the program's Achilles heel relates to failures of public-public coordination. More specifically, the public leaders of the clusters, who are supposed to articulate the needs of the sector within the public sector, typically have lacked the authority or the backing needed to elicit collaboration from the public sector agencies responsible for delivering what is needed.

Vertical Market Interventions

This quadrant is without a doubt the most controversial. Subsidies or protection for specific sectors can lead to rent-seeking behavior by the private sector actors that benefit from these interventions, and can lead to arbitrariness or favoritism by the political and economic authorities. It is mostly interventions in this quadrant—heavy-handed subsidies, protection, state production, or directed credit to specific sectors—that have given industrial policy a bad reputation. For these reasons, policies in this quadrant should be pursued with great care.

Interventions in this quadrant may pursue very different goals. Some interventions are aimed at protecting sectors in decline or that lack current or potential comparative advantages, but have a strong lobbying presence. Rice protection in Costa Rica (see Box 2.2) is a case in point. This type of intervention, which responds to rent seeking and not to the pursuit of efficiency, should clearly be discouraged. This does not mean that governments should never become involved with declining sectors. But when policies are necessary to protect workers or installed productive capacity, this should be done in ways that facilitate the reallocation of the factors of production into more productive sectors, and not by resorting to interventions that preserve these factors in unproductive sectors with limited potential.

Box 2.2 A Tale of Two Interventions

Dysfunctional Market Interventions in Costa Rica

Rice is one of the most protected commodities in Costa Rica. Tariffs are high (35 percent) and prices are controlled at each stage of the production process. Overall, rice subsidies in Costa Rica amount to 45 percent of the domestic price, higher than those in the United States (31 percent) and the European Union (32 percent) (Monge-González, Rivera, and Rosales-Tijerino, 2010). A central player in the rice policy has been the Corporación Arrocera Nacional (CONARROZ), created in 2002 to protect producers from international price shocks and improve local production conditions. But the efforts of CONARROZ, which has strong lobbying capabilities, have focused entirely on the first objective (protection) and not the second (productivity).

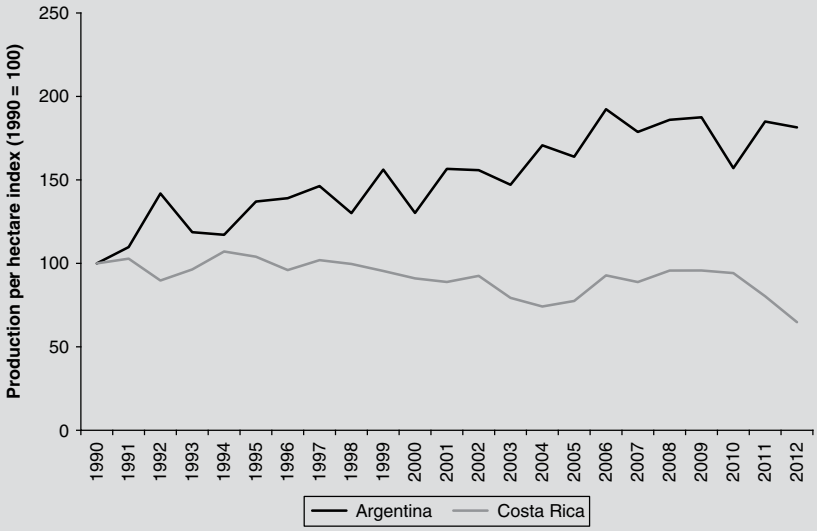
When local rice production falls below local demand (as is typically the case), import quotas are allocated to private sector actors, who can import rice without paying the corresponding tariffs. CONARROZ manages the quotas, which are assigned to rice processors in proportion to their processing capacity. Thus, processors can purchase rice at international market prices and sell the processed rice in Costa Rica at prices that reflect the high level of protection and subsidies. This generates extraordinary rents for rice processors, particularly the large ones. Altogether, rice policy in Costa Rica implies important transfers from consumers (in particular, the poor, for whom rice represents a large share of their consumption basket) to medium and large rice producers and processors.

Effective Public Inputs and Resolution of Coordination Problems in Entre Ríos, Argentina

The experience of Costa Rica contrasts with that of Entre Ríos, Argentina, where problems in the rice sector were addressed using public inputs. In this case, the key players have been the National Agricultural Technology Institute (INTA) and Pro-Arroz, a foundation of local rice producers. Until 1998, this province produced one low-quality, low-productivity variety, mainly for export to the Brazilian market. With the devaluation of the Brazilian currency in 1999, the sector lost competitiveness. Since the early 1990s, INTA had been developing a new variety of rice (Camba) of better quality and higher productivity.

Rather than lobbying for protection or subsidies, Pro-Arroz organized the producers to complement the financing of the local chapter of INTA, INTA-Concepción, by coordinating its members' contributions. Later, and at the request of Pro-Arroz, the provincial government introduced a tax on producers that went directly into financing the research activities of INTA-Concepción. This is a clear mechanism to prevent free riding, where the state helps to solve coordination problems of the private sector. Rice producers collaborated by lending their fields for the necessary experimentation with the new variety. Thanks to the successful introduction of the new variety, the sector's productivity rose rapidly, boosting its competitiveness. INTA went on to become a global leader in rice technology, and has since developed a more sophisticated variety (Puita), which has been successfully introduced in many countries, in association with BASF, the German chemical company. The results in each country in terms of productivity are very clear, as seen in Figure B2.1.

Figure B2.1 Rice Productivity in Argentina and Costa Rica, 1990–2012



Source: Authors' calculations based on FAO (2013).

Sources: For the case of Costa Rica, see Monge-González, Rivera, and Rosales-Tijerino (2010) and Cornick (2012). For the case of Entre Ríos, Argentina, see Sánchez, Butler, and Rozemberg (2011).

On the other hand, this quadrant also includes interventions designed to develop sectors with competitive potential that would not develop without them. Does it make sense to offer market interventions in these cases? While mainstream economists have long been skeptical of this type of intervention, there are theoretical arguments—and some successful experiences—for supporting industries with latent comparative advantages, even if there are short-run social costs to doing so. Naturally, in order for interventions to be warranted, the overall benefits to society must be demonstrably greater than the costs.

One particular case in which interventions in this quadrant may be justified is when they help solve coordination problems in sectors with latent comparative advantage. A typical example would be the development of a new tourism destination. Such a development might face the following problem: without an airport, it makes no sense to invest in hotels, but if there are no hotels, it makes no sense to invest in the airport. Assuming that the destination has real potential, there is a problem of multiple equilibria: a "bad" equilibrium, in which no investment is made, and a "good" one in which both the airport and the hotels are built, and tourism thrives. How can coordinated investment by all actors involved be achieved?

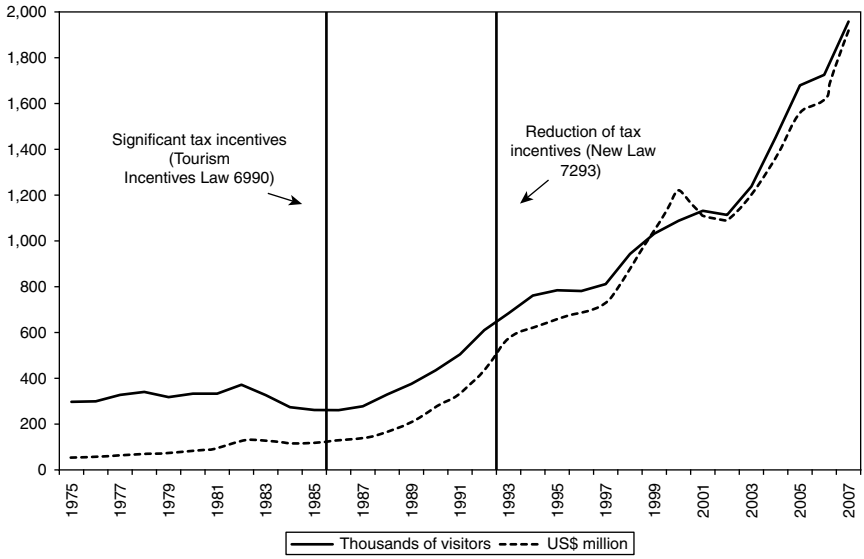
There are several ways to approach this problem. One is for the state to build the airport (an intervention in the vertical public inputs quadrant), which should provide sufficient stimulus for the investment in hotels. Another is to offer subsidies to investments in hotels and an airport, in order to coordinate these investments and achieve the good equilibrium with good outcomes. Once the tourist destination has been developed, the subsidies will no longer be necessary because the coordination problem has been solved; in any event, if required, the activity can continue to receive support through public inputs (in the quadrant discussed above). Thus interventions in this quadrant to solve coordination problems are justifiable only on a temporary basis.

A good example is Costa Rica's tourism strategy. In 1985, the country introduced a series of incentives for the sector, mostly in the form of tax breaks, covering hotel investments, air and water transport, travel agencies, and car rentals, among others. As a result

of this policy, investment in the sector took off, and the rate of arrivals and spending on tourism grew significantly. In 1992, most of these exemptions were eliminated without slowing the growth of the sector (see Figure 2.2). Later, the state assisted the sector by providing sector-specific public inputs geared toward the emergence of the sustainable tourism that the state wanted to develop. These included the creation, in collaboration with the private sector, of the national brand, "no artificial ingredients," the conservation of natural areas, and the creation of a program of sustainable tourism certification. Thus the intervention in the vertical market quadrant was temporary, and later replaced with interventions in the public inputs quadrant.

In these cases, the size and the duration of the subsidy should be the minimum necessary to resolve the coordination failure without imposing undue fiscal costs. In principle, it should be possible to coordinate the investments at little or no cost to the treasury by guaranteeing a certain return to each investor. If the project is in fact profitable provided everyone invests, everyone could reap the

Figure 2.2 Solving Coordination Failures in Tourism in Costa Rica



Source: Monge-González, Rivera, and Rosales-Tijerino (2010).

returns needed so that no actual outlays are required. The government of the province of Salta, Argentina, situated nearly 1,000 miles northwest of Buenos Aires, followed a hybrid strategy to develop its tourism sector. It offered subsidies to investment in hotels, and a minimum occupancy guarantee to a start-up airline, Andes Líneas Aéreas. At the time (2005), Salta was served by only two flights a day from Buenos Aires, and the lack of flights was considered a key obstacle for the development of the sector.[18] The government guaranteed the occupancy of 65 percent of seats in the airline, and had to transfer resources only during the first few months. A similar idea has been proposed to attract Brazilian tourists to the English-speaking Caribbean (see Box 2.3).

Box 2.3 The Caribbean-Brazil Air Bridge

Tourism flows depend on the strength of the links between the tourism destination and originating countries. The tourism sectors in the region of the CARICOM countries depend on two main markets: the large English-speaking countries in the neighborhood (the United States and Canada), and the European countries where there are strong colonial-era ties (the United Kingdom and the Netherlands for the six IDB-member countries). The United States and the United Kingdom alone concentrate nearly 60 percent of the market. The high season is from November to April when visitors come to the islands to enjoy the sand, sea, and sun.

The financial crisis of the end of the decade led to a significant drop in tourism revenue from these countries. The Caribbean countries first responded by offering hefty discounts—such as loyalty discounts or 2-for-1 deals—aimed at preserving tourists from these traditional markets. Nonetheless, the recovery has been slow and is unlikely to sustain much growth in the long term. Even traditional British and European visitors, who tend to stay for longer periods and spend more than visitors from the Western Hemisphere, are dwindling following the economic events in Europe. Therefore, the Caribbean countries have begun searching for new markets, preferably the more dynamic countries in the neighborhood.

Competition is strong and attracting tourists from new markets requires establishing a brand and airline connections. For this reason, tourism countries focus on specific markets from which they try to attract their

visitors. Having a direct connection is an obvious advantage for potential visitors, but airlines are reluctant to start a new connection without guarantees that seats will be filled. In addition, attracting tourists requires specific and substantial marketing expenses. These are two costly, up-front investments that the private sector might not be willing to make on its own.

First, attracting tourists to one Caribbean country may have significant spillovers on others, especially if visitors become interested in visiting more than one destination in a single trip, or if knowledge of one destination in the area prompts further interest in other destinations for subsequent trips. For this reason, promotion activities may be subject to a free-rider problem, if left to the private sector in each country alone. Thus, they need to be undertaken in a coordinated fashion. Second, there are important synergies between these two types of interventions. Marketing will not make a difference if connecting flights are inconvenient. And a minimum occupancy guarantee on a flight will be very expensive if the promotional efforts do not bear fruit.

Recently, the IDB conducted a pilot marketing study of outbound tourism from Brazil to analyze the merits of creating a "Caribbean-Brazil air bridge." The study, conducted in São Paulo airport (a metropolitan area of more than 20 million people), suggested that Brazilians are interested in international travel, and in many of the features that the English-speaking Caribbean has to offer. However, they are mostly unfamiliar with this region. Nonetheless, if the Caribbean region could successfully arrange frequent and direct flights from Brazil and attract just 1 percent of this potential market, tourist arrivals would increase by more than 80,000 visitors per year, a very significant number. Marketing efforts are thus essential to drive awareness and subsequently obtain greater tourism value added for the Caribbean.

While promotional activities such as marketing and advertising are not cheap, a cost-benefit analysis suggests that investing in the combined scheme would be worthwhile. The idea is that joint investment in promotional activities will address an information externality. The air bridge, in turn, would greatly enhance the returns of the promotional activities. Together, these interventions would help create a market to a destination that Brazilians did not previously know about. If so, repeat visits could eventually lead to a self-sustaining destination, at which point the minimum occupation guarantee would no longer be costly, or necessary. All this, in turn, would become a source of renewed growth for the Caribbean countries.

Aside from the resolution of coordination problems, there are other sound theoretical justifications for policies in this quadrant to help sectors achieve latent competitiveness that the market alone is unable to realize.[19] Most of them involve the "infant industry" argument. The most obvious case is that of so-called Marshallian spillovers, by which the productivity of the sector increases with its size. Since increase in size takes time, true or "latent" productivity of a certain sector may be revealed only by a fully mature industry. "Learning by doing" has often been invoked as a rationale for industrial policy: while an industry may be uncompetitive early in its life cycle, it may become competitive over time as management and workers learn to do their jobs more productively. However, individual firms have the incentive to wait until the industry is mature and profitable rather than investing in starting a process from which later entrants will benefit. Subsidies or protection may help spur the early entrants.

The benefit of industry maturation may disseminate through backward and forward linkages to other sectors of the economy. If so, the case can be made that the growth of a "strategic" sector that sets in motion this positive chain of events is worth fostering with industrial policy. Chapter 9 examines the strategic value of certain directions of productive transformation in detail. It presents a methodology to identify good candidates for a thorough analysis of whether their strategic value may merit some of the policies in this quadrant.

Melo and Rodríguez-Clare (2006) have labeled interventions like these "strategy-driven" policies—as opposed to "demand-driven" or "bottom-up" policies that respond to a sector's needs—and pose the greatest risk in terms of capture. These interventions involve governments making bets on developing a specific set of industries, many times from scratch, through a combination of aggressive market intervention policies to jump-start the sectors. Accordingly, they require a tested institutional setup that ensures competence and transparency in decision making.

Defensive PDPs

The PDPs considered so far and analyzed in depth in the following chapters are "constructive": they seek to repair the failure of the

market to deploy its capabilities in an effective way. However, building up is not all in productive development: the gains achieved must also be defended from threats. Adverse shocks that put development gains at risk include episodes of macroeconomic instability, financial crises, and patches of currency overvaluation due to extraordinary capital inflows or commodity price booms, as well as more permanent shocks such as the loss of competitiveness of a specific sector for irreversible technological reasons. The region has ample experience with the costly effects of these shocks on productive development, which at least in the case of large macroeconomic crises have caused permanent productivity losses (Blyde, Daude, and Fernández-Arias, 2010). "Defensive" PDPs to safeguard productive development from backsliding are an important complement to "constructive" PDPs in the overall policy portfolio.

While constructive PDPs aim at helping the market achieve what was latent in its capabilities (as in establishing a new competitive export activity), defensive PDPs aim at avoiding the loss of capabilities (as in closing down a competitive export activity or dismantling specialized equipment). To the extent that the cost of defensive PDPs is less than reconstructing capabilities once circumstances normalize, these PDPs are part and parcel of the formula to foster productive development. This cost/benefit test is critical: defensive PDPs only make sense when valuable productive capabilities are at risk of being destroyed. Temporary shocks fit this description only if they would destroy productivity, not simply because they cause financial pain to firms. Permanent shocks may also fit this description but only when policy is needed to soundly reallocate resources to other sectors of the economy, not to keep alive nonviable sectors.

A defensive PDP should be disciplined by the same three tests applied to constructive ones: (i) is it justified by a market failure? (ii) does the policy solution fit the problem? and (iii) are enabling institutional capabilities sufficient? Presumably it is in a firm's self-interest to preserve valuable capabilities during a temporary downturn or to reallocate its resources to the best alternative use by selling if a permanent shock makes them nonviable. However, it may fail. For example, viable firms facing a temporary downturn may lack the financial resources (and access to credit) to make the right business decisions to stay in business. If so, a suitable policy could be official

lending through development banks; this credit needs to be medium term to bridge the problem but not at concessional terms. Official credit to Brazilian exporters by BNDES during the global crisis of 2009 is an example. Other policies may address the root of the problems more directly. For example, the volatility of the real exchange rate, particularly during episodes of overvaluation, a typical shock in the region, is a serious threat to export sectors; new sectors or those that rely on supporting policies are key for productive development but may be among the most fragile. An exchange rate policy with an eye to productive development would lead to more stability and mitigate the problem. PDPs fostering the use of insurance products to cover some of these risks (such as currency risk) would be perhaps a more orderly way to address them if feasible.

Defensive PDPs would generally take the form of market interventions to isolate firms from shocks. One key aspect to consider is whether to resort to selective, vertical policies. Since policy should help those at the highest risk of failure but leave aside those able to fend for themselves (and those taking excessive risk waiting for a bailout), there is an argument to be made in favor of focalized vertical policies. Since the key driver is the destruction of valuable productive capabilities, the merits for intervention are bound to be sector specific.[20] On the other hand, it is difficult to pick the best candidates, especially under urgent circumstances. Furthermore, these defensive PDPs delivering financial support selectively would be extremely open to the risk of capture. While the benefits from defensive PDPs are potentially enormous under certain circumstances, they require substantial institutional capabilities to be successful.

The Road Map for This Book

The rest of the book is organized as follows: Part II (chapters 3–9) deals with the question of design and implementation of PDPs in different areas. Each of the seven chapters covers policies in a specific subject area, such as innovation and diffusion, financing, or internationalization policies. Rather than assuming that policies are justified, each of them starts by making or questioning the case for policy intervention, along the lines discussed in this conceptual

framework. Each presents the fundamental trends seen around the world and in the region in terms of policy design and implementation. Each then focuses on a few specific interventions from the region and elsewhere—some successful and some not—that illustrate the policy issues and may offer important policy lessons. Several chapters also discuss the results of impact evaluations, some of them performed for this book. The hope is that each chapter will inform policymakers in the region, as well as staff of the IDB on the ground, to design and implement better policies.

While the chapters are organized by subject for ease of exposition, in practice quite a bit of connection exists between policies in the different areas. This is most obvious in the case of CDPs, which often involve the identification of obstacles and opportunities for the development of a sector or value chain, and a collection of specific interventions in different areas—innovation, training, financing, internationalization—to address them. Policies addressing failures in the credit system analyzed in chapter 6 are also natural complements of policies that aim to expand investment opportunities, but require a responsive supply of credit to make a difference. Some points of connection between different areas of policy are presented throughout the book, highlighting issues such as the extent to which innovation and export promotion policies complement each other; the successful internationalization efforts of certain clusters in Argentina; or financing programs for innovation activities.

While part II focuses mostly on *what* policies to pursue, part III (chapters 10 and 11) emphasizes *how* to deliver them. The focus of part III is on the institutions for PDPs. Chapter 10 centers on the capabilities required in the public sector in order to appropriately design and implement PDPs, highlighting the overall organizational structure to coordinate its actions as well as the technical, operational, and political capabilities of productive development agencies. In addition to discussing how existing capabilities may constrain the type of policies that countries can implement effectively, the chapter explains how countries can go about building their capabilities. Great emphasis is placed on the need to encourage a learning environment in which there is room for policy experimentation and adjustment over time.

Modern PDPs have become less of a top-down affair, and increasingly involve public-private collaboration in both policy design and implementation. This collaboration is key, as the private sector has information about the sector's challenges and opportunities that is critical for effective policymaking. At the same time, if public-private interaction is not structured appropriately, it can open the door for the type of rent-seeking activities that derailed industrial policies in the past. Chapter 11 discusses in detail how to effectively structure this collaboration, using several examples from the region.